

meltingpot.io Product Requirements Document

Version 1.0 MVP PRD. Meltingpot is a collaborative coursework platform where students build a shared knowledge base around their classes. It combines LMS imports, student-created resources, collaboration tools, and practical AI assistance.

1. Product Vision

Students currently store knowledge across scattered documents, group chats, learning platforms, and personal notes. Meltingpot brings these pieces together into a shared course workspace where students can contribute, improve, and learn from each other.

2. Target Users

Primary users are students taking the same course. Secondary users include student maintainers who organize course spaces and future students who benefit from accumulated course knowledge.

3. Core Product Principles

Student-created content comes first. AI supports students instead of replacing their work. Official course material remains separate from student contributions. Collaboration should feel familiar to open source communities while remaining simple for students.

4. Main User Flow

1. Student connects Canvas or Google Classroom.
2. Meltingpot imports courses, modules, lessons, assignments, files, quizzes, tests, and deadlines.
3. Student enters a course workspace.
4. Student uploads notes or resources.
5. AI helps organize and improve material.
6. Student shares content or contributes changes to the class workspace.

5. Information Structure

The main structure is Course -> Module -> Lesson.

Each lesson contains:

- Official course content from LMS sources
- Student notes and explanations
- Resources and files
- Practice questions
- Contribution history

6. Dashboard Requirements

The dashboard should provide a quick view of a student's academic activity.

Required sections:

- Upcoming deadlines
- Enrolled courses
- Course setup
- Upload content
- Search
- Recently used items
- Streaks and contribution activity

7. AI Requirements

AI is a core part of the experience.

Required AI features:

- Clean and organize messy notes
- Turn uploads into structured study material
- Explain difficult concepts using course material
- Find gaps between notes and course resources
- Create editable practice questions
- Catch students up on missed lessons
- Summarize contribution changes for maintainers

AI-generated changes must be reviewed before becoming shared course content.

8. Collaboration Requirements

Meltingpot uses a contribution model inspired by open source workflows.

Required features:

- Copy and fork content with source attribution
- Version history
- Comments
- Branches
- Change requests
- Maintainer permissions
- Contributor history

9. Permissions and Access

Course creators manage initial permissions and can add student maintainers. Courses can be joined through email invitations, share links, or generated access codes.

10. Calendar and Search

The shared calendar combines assignments, quizzes, tests, and deadlines. Search should work across lessons, notes, files, resources, and activity.

11. Quizzes and Practice

Students create the main question bank. AI can help generate drafts, improve questions, or create practice sets from course material. Students review content before sharing.

12. Contribution Identity

Students should see their impact through a Meltingpot contribution graph, streaks, ranks, contributor profiles, and milestones. Recognition should reward meaningful contributions rather than simple activity.

13. MVP Success Criteria

A student should be able to connect a course, access class material, upload notes, improve them with AI, collaborate with classmates, submit changes, and see their contribution history.

14. Out of Scope for MVP

Deferred features include advanced certificates, adaptive learning systems, school administration tools, full semester recap exports, and deeper automation.